



Anderson acceleration for PWR whole-core pin-by-pin Nu-TH coupling calculation in NECP-Bamboo2.0

Sicheng Wang^a, Yunzhao Li^{a,*}, Liangzhi Cao^a, Lei Chen^b, Hongchun Wu^a

^a School of Nuclear Science and Technology, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China

^b TPO, Jiangsu Nuclear Power Corporation, Lianyungang, Jiangsu 222042, China

ARTICLE INFO

Keywords:

PWR
Pin-by-pin scheme
Subchannel thermal-hydraulics
Anderson acceleration
NECP-Bamboo2.0

ABSTRACT

For PWR whole-core pin-by-pin neutronics/thermal-hydraulics (Nu-TH) coupling analysis, the Picard iteration strategy is widely employed benefitting from the convenience of implementation. Nevertheless, it is limited to the global multi-physics convergence rate regarding its oscillatory convergence behaviors in the local solution of each single-physics field. In this paper, the Anderson Acceleration (AA) scheme is employed and investigated to enhance the convergence performance of the pin-by-pin Nu and subchannel TH coupling analysis code NECP-Bamboo2.0. Different from the classical relaxation scheme where the relaxation factor is predetermined and fixed, the AA scheme utilizes the linear combination of several previous solutions to correct the current solution. The coefficients of this linear combination can be obtained by solving the residual equation, making the approach more robust. In addition, the selection of the accelerated solution vector is investigated based on the tightly coupling strategy of Nu-TH in NECP-Bamboo2.0. It is illustrated that the neutron-transport solution vector could have superior theoretical acceleration performance compared to the commonly used TH solution vector in the practical PWR analysis scenario. Finally, the AA scheme in NECP-Bamboo2.0 is validated using the VERA#6 3D single-assembly transport/TH coupling benchmark problem and the VERA#7 3D whole-core multi-physics coupling benchmark. It is demonstrated by the numerical results that the AA scheme possesses good robustness and computational efficiency. For the 3D whole-core VERA#7 benchmark, compared with the classic Picard iteration with a relaxation factor of 0.5, the AA scheme based on the neutron-transport solution vector can reduce the number of global iterations and calculation time by around 80%.

1. Introduction

The physics analysis of Pressurized Water Reactors (PWRs) core aims to provide accurate neutronics data, which is essential for the subsequent safety analyses. With the advancements in computational technology and the growing emphasis on the accuracy of pin-wise distributions, PWR physics analysis has progressively attempted on refined neutronics analysis and detailed neutronics/thermal-hydraulics (Nu-TH) coupling (Gómez et al., 2012; Ferraro et al., 2020; Kochunas et al., 2017). Among these attempts, the pin-by-pin Nu approach (Yamamoto et al., 2003), when coupled with a subchannel TH solver (Ramos et al., 2016), is a promising technique scheme. It can eliminate the errors associated with assembly homogenization and pin-power reconstruction regarding the neutronics calculation (Rempe et al., 1989) and take the pin-wise TH feedback into account. In the pin-by-pin Nu-TH coupling, the minimum computational meshes for various

physics fields are refined to the pin-cell level, making the convergence constraint also need to be satisfied at the pin-wise scale. This also emphasizes the heterogeneous arrangement of pin-cells like control rods or burnable absorbers, which is ignored in the traditional coarse-mesh nodal Nu and single-channel TH coupling core code. Consequently, the numerical characteristics of pin-by-pin Nu-TH coupling necessitate the development of appropriate coupling strategies.

Generally, the Nu-TH coupling strategies in PWR physics analysis include the classical operator-splitting approach (Zerkak, et al., 2015), Picard iteration (or fixed-point iteration) (Ragusa and Mahadevan, 2009; Senecal and Ji, 2018), and more recently developed techniques like Jacobian-Free Newton-Krylov (JFNK) approach (Yang, et al., 2020). The classical operator-splitting approach sequentially solves the Nu and TH equations sequentially but only once within each time-step and usually involves an external data transfer, which may not adequately address the intricate coupling relationships. Picard iteration is an

* Corresponding author at: School of Nuclear Science and Technology, Xi'an Jiaotong University, 28 West Xianning Road, Xi'an, Shaanxi 710049, China.
E-mail address: yunzhao@mail.xjtu.edu.cn (Y. Li).

improvement of the operator-splitting method, overcoming the disadvantages by introducing an iterative process between different physics fields. Picard iteration enables a more rigorous consideration of the interactions between various fields, while easy to extend existing codes or solvers without significant modifications, thus widely applied in PWR Nu-TH coupling analysis. The examples of the Nu codes coupled with TH solver using this approach include the assembly-homogenized nodal code (Yang et al., 2018b), the high-fidelity deterministic code (Park et al., 2020) and Mento Carlo code (Ferraro et al., 2019). There is also the JFNK approach, which utilizes the equations of different physics fields and directly obtains a solution based on Newton's method. The implementation of the JFNK method requires extensive code restructuring to meet the prerequisites of solving the coupled multi-physics equations simultaneously, making it difficult to directly apply to existing codes (Wang et al., 2020). Considering the stability and technological maturity, mainstream pin-by-pin codes including SCOPE2, DYN3D, PARCS, and so on, chooses the Picard iteration for their Nu-TH coupling (Yamamoto and Ikeno, 2005; Daeubler et al., 2015; Basualdo et al., 2020).

Despite its advantages, Picard iteration does not offer unconditional stability and optimal convergence in the PWR-core Nu-TH coupling. It faces challenges such as local over-convergence in different physics fields. The local over-convergence occurs because the computation of each physical field depends on the boundary conditions provided by other physical fields in the Nu-TH Picard iteration. Therefore, during the global Nu-TH iteration process, if the boundary conditions provided by the other physical field deviate from the true solution, and the current physical field undergoes strict and expensive local convergence calculations, firstly, it will lead to inefficiency in the global iteration process. And considering that the converged solution of this current physical field will also deviate from the true solution, when it is used as a boundary condition for the subsequent calculation of the other physical fields, in some cases, numerical oscillations or even divergence occurs in the global solution (Senecal and Ji, 2017). To improve the efficiency and stability of Picard iteration, the introduction of a relaxation factor is a common choice (Müller, 2022). However, the selection of the relaxation factor is user-dependent and often remains constant throughout the iteration process, making this approach rely on experience. Relaxing the local convergence limit is another technique, but it cannot guarantee convergence from a theoretical standpoint. Therefore, even though these techniques could offer improvements with an insignificant modification of the code, they also present limitations that must be carefully managed in PWR-core analysis. Beyond the methods, the Anderson acceleration (AA) (Anderson, 2019), a quasi-Newton method, is a feasible optimization strategy for Picard-based pin-by-pin Nu-TH coupling. It could achieve a theoretical linear combination of solution vectors by constructing and solving a residual system with previous iteration solutions. The section of the solution vectors for linear combination also deserves attention in PWR pin-by-pin Nu/TH coupling, where the number of solution vectors increases significantly due to the refinement of the spatial mesh and the numerical magnitudes of the neutronics transport solution vectors and TH solution vectors differ. Considering the storage requirements and complexity of the numerical characteristics, the AA based on the global solution vectors combined with the neutron-transport solution vector and TH solution vector (Nu/TH-AA) is relatively expensive and needs additional normalization. Therefore, it is practically meaningful to investigate AA based solely on TH solution vectors (TH-AA) or neutron-transport solution vectors (Nu-AA) for its implementation in pin-by-pin Nu-TH coupling calculation. Existing implementations of AA on Nu-TH coupling have primarily focused on assembly-wise problems within a whole-core problem or loosely coupled scenarios, where each local TH calculation is implemented based on the converged neutron-transport/nuclide-density/criticality solution in a global iteration (Hamilton et al., 2016; Facchini et al., 2021; Lee and Joo, 2022). In these scenarios, it is illustrated that using TH-AA or Nu-AA alone is equivalent, hence the Nu/TH-AA could be simplified to

either of TH-AA or Nu-AA.

NECP-Bamboo2.0 is a two-step PWR-core pin-by-pin physics analysis code developed at the Nuclear Engineering Computational Physics (NECP) Lab. The classical Picard iteration with relaxation is employed originally under a so-called tightly coupling strategy for the whole-core calculation. This strategy allows various feedback, including the sub-channel TH module, the micro-depletion module, and the critical search module, to synchronously influence cross-section calculations (Li et al., 2020). While tightly considering the cross-sectional feedback, it also increases the computational complexity. Consequently, the classical Picard iteration with relaxation factor meets challenges in the 3D whole-core pin-by-pin calculations. In this paper, the AA was employed to improve the performance of the Nu-TH coupling in NECP-Bamboo2.0. Besides, the numerical characteristics of this tightly coupling strategy also differ from those in existing studies considering the simultaneous feedback of TH, micro-depletion and critical search calculation. As a result, the conclusion in previous work that the TH-solution-based AA is similar to the neutronics-solution-based AA does not hold unconditionally. Therefore, the impact of different accelerated vectors on the acceleration effect in practical computations was theoretically investigated and evaluated in this paper. The neutron-transport solution vector, other than the commonly used TH solution vector, is selected to be the accelerated vector in NECP-Bamboo2.0.

The rest of this paper is organized as follows. Section 2 briefly introduces NECP-Bamboo2.0, outlining its tight coupling strategy. The general strategy of AA is presented in Section 3, analyzing the suitability of different accelerated solution vector selection strategies under two different scenarios. Section 4 demonstrates the effectiveness of AA compared with the traditional relaxation scheme in the practical whole-core pin-by-pin analyses through two VERA benchmarks with various coupling scenarios.

2. NECP-Bamboo2.0 and its coupling strategy

2.1. Cross-section calculation and single-physics calculations

In the two-step pin-by-pin calculation, the lattice code conducts high-order heterogeneous 2D single-assembly calculations to fit the corresponding functional relationship between different state parameters and the pin-cell-homogenized cross-sections (Zhang et al., 2016), which is called the few-group constants data library. Based on the library and the core operation conditions, the corresponding cross-sections could be obtained in the core code for whole-core pin-by-pin calculation. The default state parameters used for this cross-section feedback consist of pin-wise burnup (BU), boron concentration (CB), moderator temperature (TM), and fuel temperature (TF) in NECP-Bamboo2.0. Usually, it also considers the contribution of important nuclides by implementing the micro-depletion (Wang et al., 2023) as shown in Eq. (1).

$$\Sigma_x^{\text{Cal}} = \Delta \Sigma_x^{\text{Lib}}(BU, CB, TM, TF) + \sum_i N_i^{\text{Cal}} \sigma_{i,x}^{\text{Lib}}(BU, CB, TM, TF) \quad (1)$$

where the subscript x stands for different kinds of few-group constants.

In the NECP-Bamboo2.0 core code, the neutron-transport calculation is based on a SP_3 solver using the Exponential Function Expansion Nodal (EFEN) method (Yang et al., 2018a). The state parameters used in the Eq. (1) are determined by the corresponding physics modules. Usually, the critical search calculation, macro-, and micro-depletion calculation are viewed as part of the generalized Nu calculation, while the sub-channel TH solver is commonly developed independently and coupled with the generalized Nu solver with external or internal data interface (Daeubler et al., 2015).

The critical boron concentration (CBC) and critical rod position are searched using a linear extrapolation method. Regarding the micro-depletion calculation, the transient or equilibrium I/Xe/Pm/Sm equations can be solved analytically, or by the inbuilt point burnup solver

based on CRAM (Chebyshev Rational Approximation Method) or TTA (Transmutation Trajectory Analysis) method based on the more complex depletion library consisting of dozens of important fission nuclides and fission products. Regarding the TH calculation, the two-phase homogeneous flow subchannel solver START developed by KAIST (Chaudri et al., 2019) is employed for the hydraulics and thermal conduction calculation.

2.2. Tightly coupling strategy

Before the introduction of the coupling strategy of NECP-Bamboo2.0, there are several points need to be indicated. First, as customary, the Nu referred to the Nu-TH coupling here consists of neutron-transport, critical search, and micro-depletion calculation. Besides, the iteration inside a single-physics solver is defined as the local iteration, and the iteration that passes through the outer multi-physics loop will be referred to as the global iteration. Accordingly, the convergence of the neutronics source term, eigenvalue, critical boron or critical rod position, and pin-wise TF and TM is defined as the Nu-TH global convergence.

In the classical pin-by-pin Nu and subchannel TH external coupling or the object-oriented internal loose coupling (Ferraro et al., 2019; García-Herranza et al., 2017), each local TH calculation is implemented based on the converged neutron-transport/nuclide-density/criticality solution in a global iteration. Different from the classical coupling approach, START has been reconstructed and embedded in the NECP-Bamboo2.0 code based on the so-called master-slave approach. The coupling of pin-by-pin Nu module and subchannel TH solver is based on a unified MPI-based parallelism strategy and a unified domain decomposition scheme to ensure a high level of global parallel efficiency, avoiding the additional data interfaces to handle the difference in geometry between the neutronics and the TH modules (Wang et al., 2024). Benefiting from this embedment of START into NECP-Bamboo2.0, the following tightly coupling strategy shown in Fig. 1 is employed, which

could consider the impact of cross-section feedback more efficiently and tightly.

In terms of the specific solving process shown in Fig. 1, the initial cross-sections are calculated first at each global iteration. With the given cross-section, the SP₃ multi-group calculation is implemented until the local neutronics outer iteration converges. After the neutron-transport locally converges, the subchannel TH calculation, critical search calculation, depletion calculation, and cross-section update are performed in sequence. The above steps will be iterated until Nu-TH globally converges.

2.3. Picard iteration and its relaxation

The basic principle of the Picard iteration is to convert the nonlinear algebraic equation $H(\mathbf{u}) = 0$, $H: \mathbf{R}^n \rightarrow \mathbf{R}^n$ into a fixed-point problem $\mathbf{u}^* = G(\mathbf{u}^*)$, $G: \mathbf{R}^n \rightarrow \mathbf{R}^n$. Here the function G is referred to as the iteration function, and the solution vector $\mathbf{u}^*$ is employed to represent the exact solution of the certain system. Based on this fixed-point equation, the iterative formula can be constructed as Eq. (2) for the solution, with a given initial solution and continuously iterating and updating the solution vector until the convergence condition is met.

$$\mathbf{u}^{k+1} = G(\mathbf{u}^k), \quad k = 0, 1, \dots \quad (2)$$

where the superscript k stands for the number of the iteration step.

In the pin-by-pin Nu-TH tightly coupling strategy as shown in Fig. 1, the solution vector is split into the combination of neutron-transport solution $\mathbf{u}_{tran.}$, moderator temperature $\mathbf{u}_{tm}$, fuel temperature $\mathbf{u}_{tf}$, CBC $\mathbf{u}_{cb}$, nuclide density $\mathbf{u}_{nd}$ as $\mathbf{u} = [\mathbf{u}_{tran.}, \mathbf{u}_{tm}, \mathbf{u}_{tf}, \mathbf{u}_{cb}, \mathbf{u}_{nd}]$. Considering that these single-physics solution vectors were not updated synchronously, hence this solution process could be abstracted into a general Gauss-Seidel Picard iteration as illustrated in Fig. 2 or the following equation:

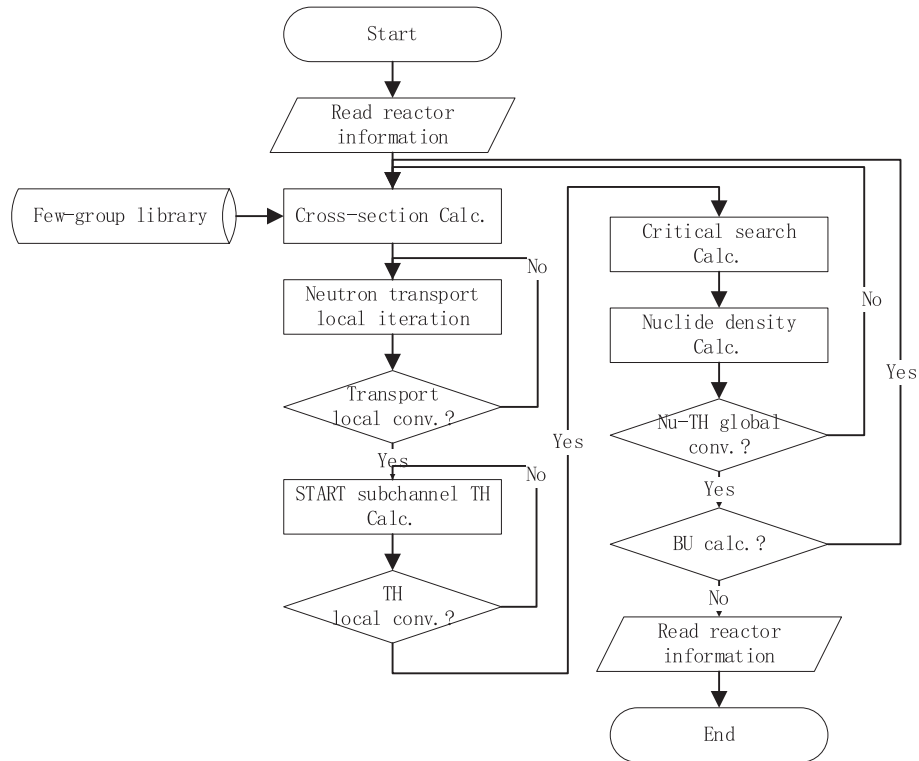


Fig. 1. Whole-core pin-by-pin Nu-TH coupling calculation flowchart.

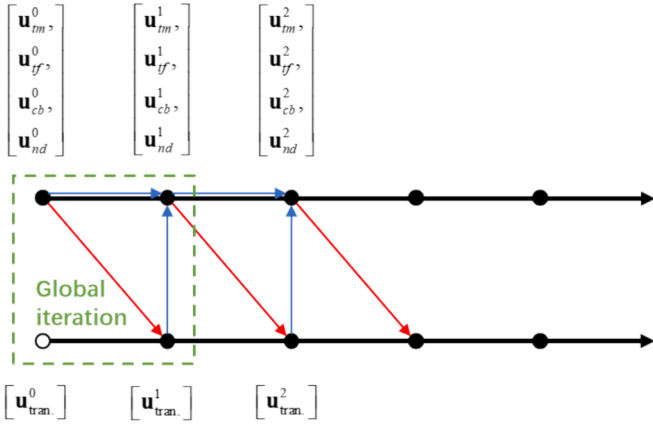


Fig. 2. Gauss-Seidel Picard iteration implemented in NECP-Bamboo2.0.

$$\begin{cases} (a) : \mathbf{u}_{\text{tran}}^{k+1} = G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k) \\ (b) : \mathbf{u}_{\text{tm}}^{k+1} = G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}), \quad \mathbf{u}_{\text{tf}}^{k+1} = G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^{k+1}) \\ (c) : \mathbf{u}_{\text{cb}}^{k+1} = G_{\text{CS}}(\mathbf{u}_{\text{tran}}^{k+1}) \\ (d) : \mathbf{u}_{\text{nd}}^{k+1} = G_{\text{ND}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k) \end{cases} \quad (3)$$

where G_{TR} is the SP_3 neutron transport operator, which could update the eigenvalue, flux, and power distribution with the given few-group constants. G_{TM} and G_{TF} are the hydraulics and thermal conduction operators, which could be combined as a TH operator G_{TH} since they are both implemented in the subchannel TH module. G_{CS} is the critical search operator, which could update the CBC or the critical rod position based on the neutronics results. In this paper, only the CBC is discussed for convenience. G_{ND} is the micro-depletion operator, considering the coupling strategy in Fig. 1 and the cross-section calculation model in Eq. (1), the microscopic cross-sections used for the micro-depletion calculation at k^{th} step are fitted with the state parameters obtained at the $(k-1)^{\text{th}}$ step. Besides, it should also be noted that in the Gauss-Seidel Picard iteration used in NECP-Bamboo2.0 as demonstrated in Fig. 2, the initial neutron transport solution $\mathbf{u}_{\text{tran}}^0$ has no influence on the global iteration, while $\mathbf{u}_{\text{tran}}^1$ is essentially the first solution vector that participates in the global iteration.

With the application of the Picard strategy, the local iteration is required for each single-physics module to satisfy its local convergence criterion. Accordingly, local over-convergence phenomena are prone to occur, and even divergence may arise in the Nu-TH iteration (Senecal and Ji, 2017). The relaxation factor is a common optimization technique used to improve the Nu-TH Picard iteration process in the core code. The key point is to combine the solution vector obtained in the current step and the previous step by using a relaxation factor ω , according to the following formula:

$$\mathbf{u}^{k+1} = \omega G(\mathbf{u}^k) + (1 - \omega)\mathbf{u}^k, \quad 0 < \omega < 1 \quad (4)$$

Essentially, the implementation of the relaxation factor could use the residual information from the previous iteration step to correct the solution in the current iteration for a negative feedback system (usually, the TH feedback in the PWR is negative). The employment of the relaxation factor is straightforward, however, the choice of the relaxation factor value (generally between 0.3 and 0.7), the strategy of the application in the operator-split scenarios (for instance, the selection of the relaxed variables) are all user-dependent, consistent in the calculation from the beginning to the end. Hence, the performance of the global

iteration is strongly related to the user's choices under the relaxation approach, making its stability difficult to ensure.

3. Anderson acceleration in NECP-Bamboo2.0

3.1. Basic theory of Anderson acceleration

Considering the drawbacks of the relaxation factor, a potentially more reliable technique route is to optimize the Picard iteration based on the adjustment of solution vectors using more information from several previous iteration steps, for example, vector extrapolation methods (Jbilou and Sadok, 2000) and AA methods. The vector extrapolation method transforms the original solution vector to a new one with better theoretical convergence properties after a sufficient number of iterations. In contrast, AA could construct a linear space using the solution vectors from the previous solutions at the very beginning of the global iteration. Therefore, it is more suitable for the pin-by-pin Nu-TH coupling, where only dozens of global iterations exist. The fundamental idea of this AA algorithm is to express the current solution as a linear combination of previous solutions.

$$\mathbf{u}^{k+1} = (1 - \beta) \sum_{i=0}^{m_k} \alpha_i^k \mathbf{u}^{k-m_k+i} + \beta \sum_{i=0}^{m_k} \alpha_i^k G(\mathbf{u}^{k-m_k+i}) \quad (5)$$

where m_k represents the number of previous solutions used for the correction at the k^{th} step, called the acceleration depth. Usually, the value of the damping factor β in the equation is set to be 1.0 (Toth and Kelley, 2015), hence Eq. (5) can be simplified into

$$\mathbf{u}^{k+1} = \sum_{i=0}^{m_k} \alpha_i^k G(\mathbf{u}^{k-m_k+i}) \quad (6)$$

Before the determination of the coefficients α_i^k in Eq. (6), the residual for each iteration and the residual matrix for the iterated system could be defined as

$$\mathbf{f}^k = G(\mathbf{u}^k) - \mathbf{u}^k \quad (7)$$

$$\mathbf{F}^k = [\mathbf{f}^{k-m_k}, \dots, \mathbf{f}^k] \quad (8)$$

Considering that the exact solution $\mathbf{u}^*$ of the fixed-point iteration problem satisfies that $\mathbf{f}^* = G(\mathbf{u}^*) - \mathbf{u}^* = 0$, the coefficients α_i^k in Eq. (6) can be determined by solving the constrained least-square optimization problem

$$\min_{\alpha^k} \|\mathbf{F}^k \alpha^k\|_2, \quad \text{s.t.} \quad \sum_{i=0}^{m_k} \alpha_i^k = 1 \quad (9)$$

However, the solution of a constrained least-square problem is relatively complex. With the following notation,

$$\begin{cases} \Delta \mathbf{u}^k = \mathbf{u}^{k+1} - \mathbf{u}^k \\ \Delta \mathbf{G}^k = G(\mathbf{u}^{k+1}) - G(\mathbf{u}^k) \\ \Delta \mathbf{f}^k = \mathbf{f}^{k+1} - \mathbf{f}^k = \Delta \mathbf{G}^k - \Delta \mathbf{u}^k \end{cases} \quad (10)$$

$$\begin{cases} \tilde{\mathbf{u}}^k = [\Delta \mathbf{u}^{k-m_k}, \dots, \Delta \mathbf{u}^k] \\ \tilde{\mathbf{G}}^k = [\Delta \mathbf{G}^{k-m_k}, \dots, \Delta \mathbf{G}^k] \\ \tilde{\mathbf{F}}^k = [\Delta \mathbf{f}^{k-m_k}, \dots, \Delta \mathbf{f}^k] \end{cases} \quad (11)$$

$$\gamma_i^k = \sum_{j=0}^i \alpha_j^k, \quad i = 0, 1, \dots, m_k - 1 \quad (12)$$

the constrained least-square problem could be transformed into an unconstrained least-square problem as

$$\min_{\gamma^k = [\gamma_0^k, \dots, \gamma_{m_k-1}^k]^T} \|\mathbf{f}^k - \tilde{\mathbf{F}}^k \gamma^k\|_2 \quad (13)$$

and the modified iterative solution can be determined as

$$\mathbf{u}^{k+1} = G(\mathbf{u}^k) - \tilde{\mathbf{G}}^k \gamma^k \quad (14)$$

In summary, the pseudo code for AA is presented in Table 1.

It can be demonstrated from Table 1 that for the general Anderson acceleration, the modification of the solution vector starts from $\mathbf{u}^2$, while $\mathbf{u}^1$ is obtained directly from the initial given value $\mathbf{u}^0$ without any modification. Regarding the acceleration depth in AA, several points need to be supplemented:

- I. When it is set to be k at each step as $m_k = k$, according to the theoretical analysis by Walker et al. (Walker and Ni, 2011), the AA has demonstrated its equivalence to the generalized minimal residual method (GMRES) (Saad and Schultz, 1986), both of which employ the idea of minimizing the residual to determine coefficients.
- II. When the acceleration depth is set to be 0 as $m_k = 0$, the AA degenerates to the original Picard iteration.
- III. When the acceleration depth is 1, and the correction coefficient is set to be a fixed value as $m_k = 1$, $\alpha_0^k = w$, $\alpha_1^k = 1 - w$, the AA degenerates to the Picard iteration with relaxation. However, compared to the actual AA with $m_k = 1$, the fixed correction coefficient in the relaxation scheme may result in insufficient correction in the early iterations and excessive correction in the later iterations.

For the convenience of description, AA with an acceleration depth of m_k will be denoted as AA- m_k in the following content.

3.2. Implementation of Anderson acceleration in NECP-Bamboo2.0

For solving the unconstrained least-square problem Eq. (13), the common implementations for small-scale problems typically rely on QR decomposition based on Householder transformation. However, the total number of spatial meshes in the pin-by-pin Nu-TH coupling is generally on the order of millions, and the numerical computations are performed based on domain decomposition in NECP-Bamboo2.0. In this case, using QR decomposition to solve the equation would lead to complex data communication and storage requirements between different processors for storing and computing the coefficient matrix.

Therefore, in the implementation of AA in this paper, a classical transposed matrix approach is adopted for solving the least-square problem. The general formula is as follows:

$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{b} \quad (15)$$

And based on the concept of block matrices, the computation of the $\mathbf{A}^T \mathbf{A}$ on the left-hand side can be expressed as

$$\begin{aligned} (\mathbf{A}^T \mathbf{A})_{m_k \times m_k} &= \begin{bmatrix} \mathbf{A}_{CPU0} \\ \mathbf{A}_{CPU1} \\ \dots \\ \mathbf{A}_{CPUn} \end{bmatrix}^T \begin{bmatrix} \mathbf{A}_{CPU0} \\ \mathbf{A}_{CPU1} \\ \dots \\ \mathbf{A}_{CPUn} \end{bmatrix} \\ &= [\mathbf{A}_{CPU0}^T, \mathbf{A}_{CPU1}^T, \dots, \mathbf{A}_{CPUn}^T] \begin{bmatrix} \mathbf{A}_{CPU0} \\ \mathbf{A}_{CPU1} \\ \dots \\ \mathbf{A}_{CPUn} \end{bmatrix} = \sum_{i=0}^n \mathbf{A}_{CPUi}^T \mathbf{A}_{CPUi} \end{aligned} \quad (16)$$

Based on the formula, It is only necessary to calculate the local $\mathbf{A}_{CPUi}^T \mathbf{A}_{CPUi}$ on each CPU first and then perform a simple matrix addition with the MPI function. Similarly, the right-hand vector $\mathbf{A}_{CPUi}^T \mathbf{b}_{CPUi}$ can be calculated locally, followed by a simple vector addition. Considering that the acceleration depth of AA generally does not exceed 10 to

Table 1

Pseudo code for general Anderson acceleration.

Given $\mathbf{u}^0$ and m_k
Set $\mathbf{u}^1 = G(\mathbf{u}^0)$
For $k = 1, 2, \dots$ Until converge, do
Set $m^k = \min(m, k)$
Calculate $G(\mathbf{u}^k)$
Set $\mathbf{f}^k = G(\mathbf{u}^k) - \mathbf{u}^k$
Set $\tilde{\mathbf{F}}^k$ and $\tilde{\mathbf{G}}^k$ according to Eq.
Determine $\gamma^k = [\gamma_0^k, \dots, \gamma_{m^k-1}^k]$, which solves $\min_{\gamma^k} \|\mathbf{f}^k - \tilde{\mathbf{F}}^k \gamma^k\|_2$
Set $\mathbf{u}^{k+1} = G(\mathbf{u}^k) - \tilde{\mathbf{G}}^k \gamma^k$
End for

achieve satisfactory acceleration effects, the rank of $\mathbf{A}_{CPUi}^T \mathbf{A}_{CPUi}$ cannot exceed 10, accordingly. Consequently, the communication cost between different CPUs is low, allowing for efficient handling of coefficient calculations.

3.3. Selection of the accelerated solution vector

In section 3.1, the connection and the differences between AA and the relaxation have been preliminarily explained. Considering that the relaxation approach can be informally considered as a special case of AA-1, the specific implementation process of AA also involves a similar issue as the relaxation approach related to the selection of acceleration strategies in the coupling process. This section discusses the selection strategies for the acceleration vector in two common coupling scenarios based on the tightly coupled strategy of Picard iteration in NECP-Bamboo2.0. One scenario is the simplified transport/TH coupling, and the other is the common transport/TH/criticality/micro-depletion coupling scenario faced in the practical calculation.

3.3.1. Transport/TH coupling scenario

In the transport/TH coupling scenario, the Picard iteration system Eq. (3) could be simplified to

$$\begin{cases} (a) : \mathbf{u}_{\text{tran}}^{k+1} = G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k) \\ (b) : \mathbf{u}_{\text{tm}}^{k+1} = G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}) \quad \mathbf{u}_{\text{tf}}^{k+1} = G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^{k+1}) \end{cases} \quad (17)$$

Eq. (17) could also represent the loosely coupling strategy as mentioned in Section 1, since in the loosely coupling strategy only the combined neutronics solution of the converged transport/nuclide-density/criticality is necessary to transfer to the TH module. For this coupling scenario, the neutron-transport calculation at the $(k+1)^{\text{th}}$ global iteration step is performed firstly as Eq. (17)-(a). Subsequently, TH calculations are conducted and could be transformed with a combined TH operator as:

$$\begin{aligned} \begin{bmatrix} \mathbf{u}_{\text{tm}}^{k+1} \\ \mathbf{u}_{\text{tf}}^{k+1} \end{bmatrix} &= G_{\text{TH}}(\mathbf{u}_{\text{tran}}^{k+1}) = \begin{bmatrix} G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}) \\ G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^{k+1}) \end{bmatrix} = \begin{bmatrix} G_{\text{TM}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k)) \\ G_{\text{TF}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k), \mathbf{u}_{\text{tm}}^{k+1}) \end{bmatrix} \\ &= \begin{bmatrix} G_{\text{TM}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k)) \\ G_{\text{TF}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k), G_{\text{TM}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k))) \end{bmatrix} \end{aligned} \quad (18)$$

After the $(k+1)^{\text{th}}$ global iteration step is completed, repeat the iterative process and the neutron-transport solution vector for the $(k+2)^{\text{th}}$ step could be obtained and transformed as

$$\mathbf{u}_{\text{tran}}^{k+2} = G_{\text{TR}}(\mathbf{u}_{\text{tm}}^{k+1}, \mathbf{u}_{\text{tf}}^{k+1}) = G_{\text{TR}}(G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}), G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}))) \quad (19)$$

It could be illustrated from Eq. (19) that the $(k+1)^{\text{th}}$ neutron-transport

solution vector is the function of the k^{th} neutron-transport solution vector for the transport/TH coupling system from the step $k = 1$. Similarly, the $(k+1)^{\text{th}}$ TH solution vector is the function of the k^{th} TH solution vector according to Eq. (18) from the step $k = 0$. Given the Gauss-Seidel Picard iteration model illustrated in Fig. 2, both the neutron-transport solution vector and the TH solution vector can independently construct the AA system as shown in Table 1. The difference lies in the fact that, in the symbol system used in this paper as demonstrated in Fig. 2 and Eq. (3), the AA based on TH solution vector (TH-AA) can be implemented starting from $(\mathbf{u}_{\text{tm}}^2, \mathbf{u}_{\text{tf}}^2)$, while the first modified solution vector for AA based on neutron-transport solution vector (Nu-AA) is $(\mathbf{u}_{\text{tran}}^3)$.

Therefore, in the Transport/TH coupling system, acceleration can be applied to the combined solution vector $(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k)$, $k = 0, 1, \dots$ consisting of the neutron-transport solution vector and the TH solution vector from the perspective of the global iteration, or the separate and independent neutron-transport solution vector $(\mathbf{u}_{\text{tran}}^{k+1})$, $k = 0, 1, \dots$ or TH solution vector $(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k)$, $k = 0, 1, \dots$. Considering the numerical differences between the neutron-transport and TH solution vectors, when adopting the combined one for the acceleration, additional mathematical normalization is required, making the acceleration based on the combined solution vector expensive from the perspective of code development and storage. Hence, a separated solution vector is more practical for the pin-by-pin calculation. Besides, under this scenario, the transport and TH solution vectors are equivalent, which means the effectiveness of AA based on the transport solution vector and on the TH solution vector should be at the same level.

3.3.2. Transport/TH/criticality/micro-depletion coupling scenario

Considering the tightly coupling strategy in the practical calculation of NECP-Bamboo2.0, the transport/TH/criticality/micro-depletion coupling scenario is analyzed as follows. For this coupling scenario, the neutron-transport calculation at the $(k+1)^{\text{th}}$ global iteration step is performed firstly as Eq. (3)-(a). Subsequently, TH calculations are conducted and could be transformed with a combined TH operator as

$$\begin{aligned} \begin{bmatrix} \mathbf{u}_{\text{tm}}^{k+1} \\ \mathbf{u}_{\text{tf}}^{k+1} \end{bmatrix} &= G_{\text{TH}}(\mathbf{u}_{\text{tran}}^{k+1}) = \begin{bmatrix} G_{\text{TM}}(\mathbf{u}_{\text{tran}}^k) \\ G_{\text{TF}}(\mathbf{u}_{\text{tran}}^k, \mathbf{u}_{\text{tm}}^{k+1}) \end{bmatrix} = \begin{bmatrix} G_{\text{TM}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k)) \\ G_{\text{TF}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k), \mathbf{u}_{\text{tm}}^{k+1}) \end{bmatrix} \\ &= \begin{bmatrix} G_{\text{TM}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k)) \\ G_{\text{TF}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k), G_{\text{TM}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k))) \end{bmatrix} \end{aligned} \quad (20)$$

Following that it is the critical search calculation, as indicated by Eq. (3)-(c), which is related to the computed value of the eigenvalue. Therefore, it can also be considered as a function of either flux or other solutions,

$$\mathbf{u}_{\text{cb}}^{k+1} = G_{\text{CS}}(\mathbf{u}_{\text{tran}}^{k+1}) = G_{\text{CS}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k)) \quad (21)$$

The following is the microscopic depletion calculation. The microscopic depletion calculation used to update the nuclear number density is

essentially a point depletion calculation. The input variables required for the point depletion calculation include the flux obtained from the neutron-transport calculation, as well as the homogenized few-group microscopic cross-sections provided by the cross-section calculation model. Therefore, it can be expressed as a function of flux and microscopic cross-sections:

$$\begin{aligned} \mathbf{u}_{\text{nd}}^{k+1} &= G_{\text{ND}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k) \\ &= G_{\text{ND}}(G_{\text{TR}}(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k), \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k) \end{aligned} \quad (22)$$

There are two points that need to be noted here. Firstly, although the TH parameters and CBC have been updated before the micro-depletion calculation, the cross-section calculations have not yet been updated according to Fig. 1. Therefore, the homogenized few-group microscopic cross-sections used here are still based on the state parameters obtained from the previous global iteration. Secondly, in the core code, the macroscopic cross-sections used for the neutron transport calculation are corrected by the nuclide number densities calculated from the microscopic depletion calculation with the microscopic cross-sections obtained through back-substitution. Hence, the independent variables required for the macroscopic cross-sections in neutron-transport calculations are $(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k)$, while $(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k)$ in the micro-depletion calculation.

Compared to the coupling in Section 3.3.1, the employment of the tightly coupling strategy will significantly change the characteristic of the coupling system. Firstly, for TH calculation, it can be demonstrated from Eq. (20) that the $(k+1)^{\text{th}}$ solution vector can no longer be expressed as a function of its own converged solution vector at the k^{th} global iteration step. In other words, the TH solution vector is no longer independent. Therefore, it is no more suitable to implement the AA based solely on the TH solution vector, unless the combined solution vector as $(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k)$. However, similarly to the issues faced when using the combined solution vector $(\mathbf{u}_{\text{tran}}^k, \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k)$ in the transport/TH coupling scenario, implementing the AA based on the combined solution vector $(\mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k, \mathbf{u}_{\text{nd}}^k)$ also faces problems such as high memory demand and

the need for additional processing.

Conversely, the neutron-transport solution vector can still be represented as a function of itself, but with some changes. Similarly to Eq. (19), for this scenario, once the $(k+1)^{\text{th}}$ global iteration step is completed, repeating the iterative process and the neutron-transport solution vector for the $(k+2)^{\text{th}}$ step could be obtained and transformed as

$$\begin{aligned} \mathbf{u}_{\text{tran}}^{k+2} &= G_{\text{TR}}(\mathbf{u}_{\text{tm}}^{k+1}, \mathbf{u}_{\text{tf}}^{k+1}, \mathbf{u}_{\text{cb}}^{k+1}, \mathbf{u}_{\text{nd}}^{k+1}) \\ &= G_{\text{TR}}(G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}), G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1})), G_{\text{CS}}(\mathbf{u}_{\text{tran}}^{k+1}), G_{\text{ND}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k)) \end{aligned} \quad (23)$$

Since the microscopic cross-sections used in the $(k+1)^{\text{th}}$ are based on the state parameters from the previous iteration step as shown in Eq. (23) and $\mathbf{u}_{\text{tran}}^1$ is essentially the first solution vector that participates in the global iteration, the neutron-transport solution vector can only be expressed as a function of itself the $\mathbf{u}_{\text{tran}}^3$, with the solutions from the previous two iteration steps as follows:

$$\begin{aligned} \mathbf{u}_{\text{tran}}^{k+2} &= G_{\text{TR}}(G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}), G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1})), G_{\text{CS}}(\mathbf{u}_{\text{tran}}^{k+1}), G_{\text{ND}}(\mathbf{u}_{\text{tran}}^{k+1}, \mathbf{u}_{\text{tm}}^k, \mathbf{u}_{\text{tf}}^k, \mathbf{u}_{\text{cb}}^k)) \\ &= G_{\text{TR}} \left(\begin{bmatrix} G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1}) \\ G_{\text{TF}}(\mathbf{u}_{\text{tran}}^{k+1}, G_{\text{TM}}(\mathbf{u}_{\text{tran}}^{k+1})) \\ G_{\text{CS}}(\mathbf{u}_{\text{tran}}^{k+1}) \\ G_{\text{ND}}(\mathbf{u}_{\text{tran}}^{k+1}, G_{\text{TM}}(\mathbf{u}_{\text{tran}}^k), G_{\text{TF}}(\mathbf{u}_{\text{tran}}^k, G_{\text{TM}}(\mathbf{u}_{\text{tran}}^k)), G_{\text{CS}}(\mathbf{u}_{\text{tran}}^k)) \end{bmatrix}^T \right) \end{aligned} \quad (24)$$

Therefore, for this type of coupling scenario in the NECP-Bamboo tightly coupled strategy, although the neutron-transport solution vector can still independently construct the AA system, strictly speaking, it is a multi-dimensional fixed-point iteration. Consequently, AA system should be constructed based on a combined neutron-transport solution vector needs as $(\mathbf{u}_{\text{tran}}^{k+2}, \mathbf{u}_{\text{tran}}^{k+1})$, $k = 0, 1, \dots$, while the first modified solution vector is $(\mathbf{u}_{\text{tran}}^4, \mathbf{u}_{\text{tran}}^3)$.

However, the storage requirement for the combined neutron-transport solution vector is considerable in the pin-by-pin calculation, and the influence of this multi-dimensional formula will not be significant in the later stages of the global iteration when the solutions between the previous step and the next step are already relatively close. Hence in this paper, while adopting the combined neutron-transport solution vector approach for the transport/TH/criticality/micro-depletion coupling scenario, an approximate solution is also investigated. That is, similar to the Nu-AA strategy in the transport/TH coupling scenario demonstrated in section 3.3.1, only the single neutron-transport solution vector $(\mathbf{u}_{\text{tran}}^{k+1})$, $k = 0, 1, \dots$ is used for the AA calculation, and the modification of the solution vector starts from $(\mathbf{u}_{\text{tran}}^3)$. Numerical results later will also demonstrate that in the practical core physics analysis calculations, this approximation is acceptable.

4. Numerical results

In this section, the VERA#6 3D single-assembly benchmark, which satisfies the transport/TH scenario, and VERA#7 3D whole-core benchmark which satisfies the transport/TH/criticality/micro-depletion scenario are simulated to validate and evaluate the performance of the AA with different accelerated solution vectors. Comparisons are also made with the calculation results of the original Picard iteration with relaxation.

4.1. VERA benchmark problem 6

The VERA Core Physics Benchmark Problem (Montgomery, 2011) provides a series of problems that originally came from a PWR-core designed by Westinghouse in the United States, which is called WBN1

(Watts Bar Nuclear 1). Each fuel assembly in it contains 264 fuel rods, 1 measuring tube, and 24 guide tubes with a center-to-center spacing of 21.50 cm and a half-width of the water gap of 0.04 cm. Among this series, VERA#6 is a 3D PWR single fuel assembly with axial reflectors and two types of spacer grids, consisting of 6 mixing vanes. The axial arrangement of this benchmark problem is illustrated in Fig. 3. This problem considers the effects of TH feedback at zero-burnup HFP con-

ditions based on a given boron concentration, without considering the influence of fission products Xe/Sm. Hence, this benchmark meets the requirements of a simple transport/TH scenario.

For this benchmark, calculations are performed using original Picard iteration, relaxation, Nu-AA, and TH-AA. The relaxation is also applied based on the flux, treated the same as the Nu-AA. In the global iteration, the converged solution needs to simultaneously satisfy the convergence criterion for the eigenvalue, the 2-norm and infinity-norm of the fission source, and the infinity-norm of the temperatures, as follows:

$$\left| 1 - \frac{\text{eigenvalue}^{k+1}}{\text{eigenvalue}^k} \right| < 1 \times 10^{-6} \quad (25)$$

$$\frac{\|\psi_i^{k+1} - \psi_i^k\|_2}{\|\psi_i^k\|_2} = \frac{\sqrt{\sum_{i \in I} (\psi_i^{k+1} - \psi_i^k)^2}}{\sqrt{\sum_{i \in I} (\psi_i^k)^2}} < 1 \times 10^{-4} \quad (26)$$

$$\left\| \frac{\psi_i^{k+1} - \psi_i^k}{\psi_i^k} \right\|_{\infty} = \max_{i \in I} \left| \frac{\psi_i^{k+1} - \psi_i^k}{\psi_i^k} \right| < 1 \times 10^{-4} \quad (27)$$

$$\begin{aligned} \left\| \frac{TF_i^{k+1} - TF_i^k}{TF_i^k} \right\|_{\infty} &= \max_{i \in I} \left| \frac{TF_i^{k+1} - TF_i^k}{TF_i^k} \right| < 1 \times 10^{-4} \\ \left\| \frac{TM_i^{k+1} - TM_i^k}{TM_i^k} \right\|_{\infty} &= \max_{i \in I} \left| \frac{TM_i^{k+1} - TM_i^k}{TM_i^k} \right| < 1 \times 10^{-4} \end{aligned} \quad (28)$$

The maximum number of fission source iterations was set to 1000. The iteration results under different strategies are shown in Table 2.

From Table 2, it can be illustrated for the simple transport/TH coupling scenario, the solution of directly Picard iteration without acceleration requires a large number of global iterations. The local neutron-transport iteration is the most time-consuming part of the calculation. Considering that each global iteration involves 3 to 4 local neutron-transport iterations, a total number of approximately 700 fission source updates are needed with the calculation time of about 294 s. The relaxation scheme can accelerate global convergence, but its effectiveness depends on the user-defined relaxation factor, leading to

significant variations in global iteration counts varying the value of the relaxation factor. The best performance is the calculation with a relaxation factor as 0.7, the number of global iterations is 12, while the calculation time is about 18 s for this case. Both the Nu-AA and TH-AA demonstrate effective acceleration for the coupled transport/TH scenario. Additionally, with an increase in the value of acceleration depth, the number of global iterations will gradually converge to a constant value as 8 and 6 for Nu-AA and TH-AA, respectively. It should be noted that the slight differences in the number of global and local iterations varying acceleration depths are acceptable. The calculation time of Nu-AA and TH-AA for this scenario under a sufficient acceleration depth is 13 s and 11 s, respectively.

To elucidate the iterative characteristics of this system, firstly, the variation of k_{eff} corresponding to each fission source iteration for the original Picard iteration is depicted in Fig. 4. It could be demonstrated that the eigenvalue fluctuates with the fission source iteration process for the original Picard iteration, which is aligning with the expectations for a coupled transport/TH system with negative TH feedback due to the local over-convergency. Besides, an asymmetry of the k_{eff} bias to the converged value in the early iteration steps could also be observed. This asymmetry is attributed to the operator-splitting treatment in Picard iteration. In the early iteration steps, the TH calculation will also locally converge to a solution that will significantly deviate from the exact TH solution. Therefore, it will also affect the neutron-transport calculations,

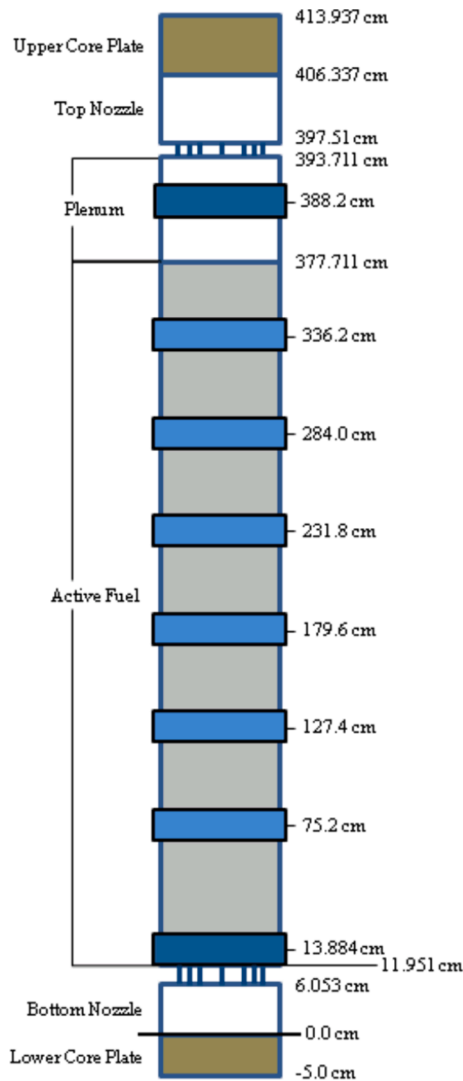


Fig. 3. Axial arrangement of the VERA benchmark problem.

Table 2

Iteration performance for VERA#6 benchmark.

		No. of global iterations	No. of transport fission source iterations	Calculation time /s
Original		222	719	294.134
Relaxation	factor =	34	139	48.615
	0.3			
	0.5	19	85	29.599
	0.7	12	56	18.179
Nu-AA	$m_k = 1$	8	38	13.634
	2	7	35	11.802
	4	8	37	13.129
	6	8	37	13.577
	8	8	37	13.263
TH-AA	$m_k = 1$	8	38	13.678
	2	6	32	11.290
	4	6	32	11.347

making the locally converged neutron-transport solution asymmetric compared to the exact solution. After sufficient global iterations in this transport/TH coupling case, the deviation of the TH solution is no longer so substantial, exhibiting the convergence process of a typical negative feedback system.

Usually, the traditional relaxation approach with a relaxation factor of 0.5, is more suitable for scenarios with symmetrical fluctuation amplitudes. The asymmetry illustrated in Fig. 4 can also explain that in the calculations for this case, the acceleration effect of the traditional approach as setting the relaxation factor to 0.5 is not as evident as when the factor is set to 0.7.

To further illustrate the acceleration effectiveness of different methods for this case, Fig. 5 presents the curves of eigenvalue variation with fission source iteration for different acceleration methods. It should be noted that as the first global iteration is based on the same assumed initial values, hence the first global iteration process is the same for different methods. Consequently, the starting point of the curves in Fig. 5 was selected to be the second global iteration (in this case, starting from the 11th fission source iteration). For the convenience of the comparison, the results with different accelerated solution vectors of AA are depicted with a depth of 1. Compared to the relaxation, the AA can be proved to be more stable, capable of capturing solution characteristics better, and achieving faster convergence. Moreover, for problems like VERA#6 involving transport/TH coupling, the Nu-AA and TH-AA exhibit nearly identical eigenvalue convergence curves, indicating a certain equivalence in acceleration effectiveness, aligning with the theoretical expectations in Section 3.3.1. Besides, the correction coefficient values during the iterations for the various AA schemes with $m_k = 1$ for the VERA#6 benchmark are illustrated in Fig. 6. The correction coefficient values of Nu-AA-1 and TH-AA-1 change in the almost same range with (0, 0.6) and could be adjusted during the iteration, which leads to the more efficient acceleration practically compared to the relaxation approach.

Regarding the convergence criterion for the eigenvalue is 10^{-6} , the eigenvalue results of different acceleration methods are consistent after retaining 5 decimal places with the eigenvalue result obtained without acceleration. Similarly, the average outlet coolant temperatures are also consistent after retaining 2 decimal places. Table 3 presents the comparison results of NECP-Bamboo2.0 with the results of some reference codes. The results of the MPACT/CTF provided by Brian N. Aviles (Aviles et al., 2016) are used as the reference solution for this study. Besides, results of the MC21/COBRA-IE and MC21/CTF provided by Kelly (Daniel et al., 2017) are also listed. The eigenvalue and the average outlet coolant temperature result demonstrate that the coupling transport/TH calculation of NECP-Bamboo2.0 has high accuracy. Besides, the RMS and maximum bias of pin-power distribution is 0.26 % and -0.48 %, respectively, also illustrating that the accuracy of pin-by-pin Nu-TH coupling NECP-Bamboo2.0 at the same level as the results of the

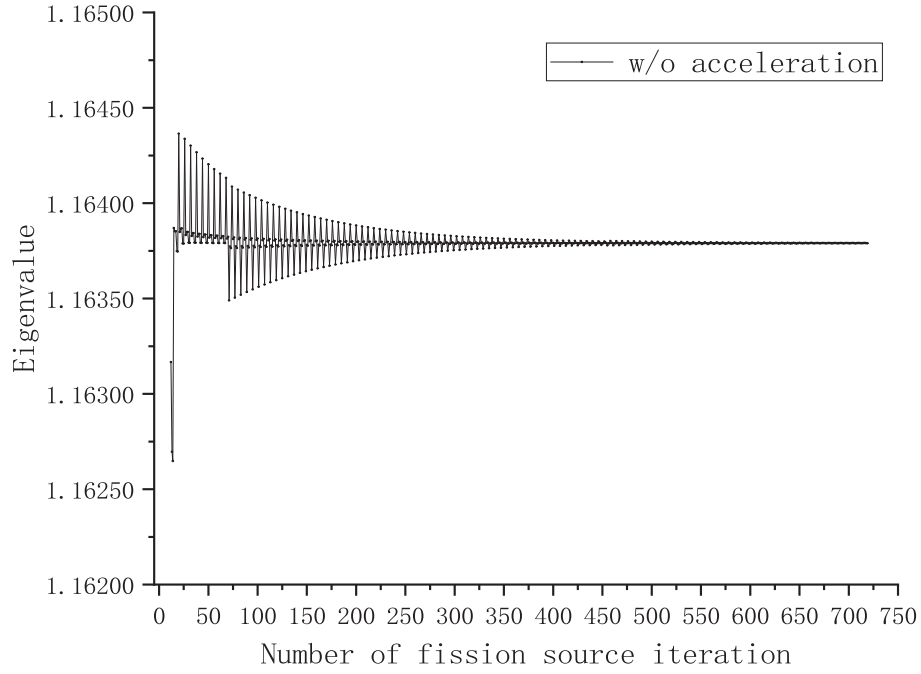


Fig. 4. k_{eff} varying fission source iteration without acceleration for VERA#6 benchmark.

referenced high-fidelity codes.

4.2. VERA benchmark problem 7

In practical reactor physics analysis, the coupled transport/TH/criticality/micro-depletion is more common scenario. In this section, the calculations based on different Picard iteration acceleration methods for the VERA#7 benchmark problem are implemented and analyzed. The VERA#7 benchmark is a 3D whole-core criticality problem with hot full power. Even though this benchmark is based on a zero-burnup case, the micro-depletion of equilibrium xenon is also considered. The axial arrangement of the VERA#7 benchmark is consistent with VERA#6, and

the radial arrangement is shown in Fig. 7. The operational parameters for VERA#7 are detailed in Table 4, with 215 steps for the withdrawal of control rod group D, while the remaining control rod groups are fully withdrawn.

For this 3D whole-core VERA#7 problem, calculations are performed using the original Picard iteration, relaxation, Nu-AA, and TH-AA. It should be noted that in the TH-AA, only the pure TH solution vector $(\mathbf{u}_{tm}^k, \mathbf{u}_{tf}^k)$ other than the combined solution vector $(\mathbf{u}_{tm}^k, \mathbf{u}_{tf}^k, \mathbf{u}_{cb}^k, \mathbf{u}_{nd}^k)$ is employed in order to validate the analysis in section 3.3.2. The pin-wise multi-group flux is selected here as the accelerated solution vector for both Nu-AA and relaxation. Additionally, the AA based on the combined vector $(\mathbf{u}_{\text{tran}}^{k+2}, \mathbf{u}_{\text{tran}}^{k+1})$, $k = 0, 1, \dots$ is also implemented in this case,

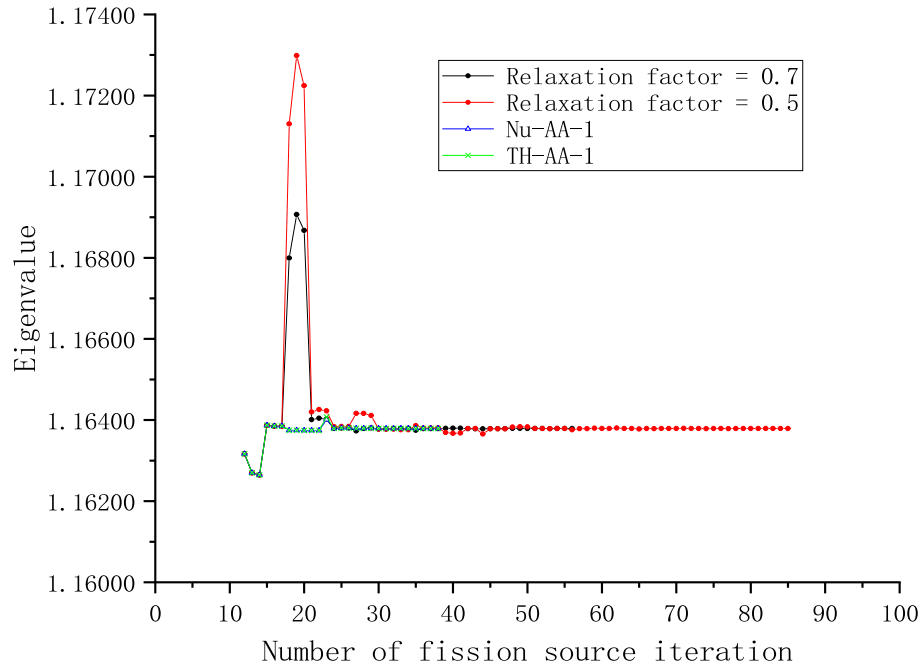


Fig. 5. k_{eff} varying fission source iteration with various acceleration methods for VERA#6 benchmark.

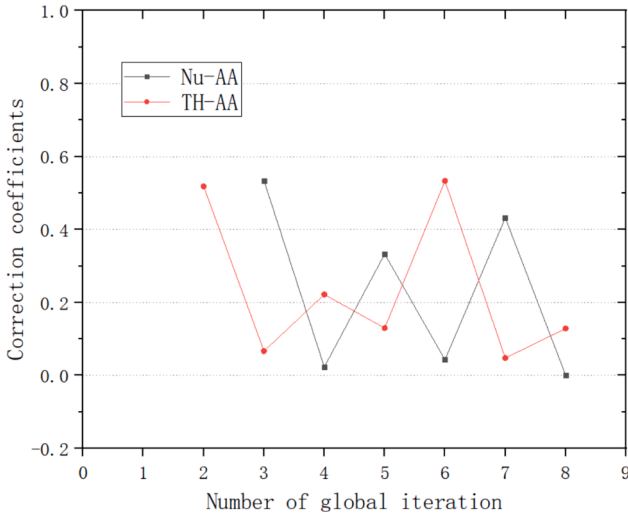


Fig. 6. Correction coefficients during the iterations for various AA schemes with $m_k = 1$ for VERA#6 benchmark.

which is referred to as Nu(CV)-AA. Accordingly, the first solution vectors to be corrected for Nu-AA and Nu(CV)-AA are (u_{tran}^3) and $(u_{\text{tran}}^4, v_{\text{tran}}^3)$, respectively. In the other words, compared to Nu-AA that the modification of the solution vector starting from $k = 3$, there is one more non-modified global iteration for Nu(CV)-AA, whose modification of the solution vector starting from $k = 4$.

The convergence criteria of flux and temperature were set the same as in the VERA#6 case. Considering that the VERA#7 problem requires the critical calculation, the criterion of eigenvalue for the critical search is set to be less than 1 pcm, while the initial boron concentration is set to be 1285 ppm in this calculation. Besides, for the calculation of this VERA#7 benchmark, 80 cores of a single AMD-PYC-7742 processor are used. The iteration performance and calculation time under different strategies are shown in Table 5.

Table 3
Comparison for VERA#6 benchmark.

Code	Eigenvalue	Average outlet coolant temperatures	RMS of pin-power distribution bias	Max. pin-power distribution bias
MPACT/CTF	1.16361 (Reference)	600.04 K	—	—
MC21/CTF	1.16424 (+63pcm)	599.72 K	—	—
MC21/COBRA-IE	1.16424 (+63pcm)	600.12 K	—	—
NECP-Bamboo2.0	1.16385 (+24pcm)	600.31 K	0.26 %	−0.48 %

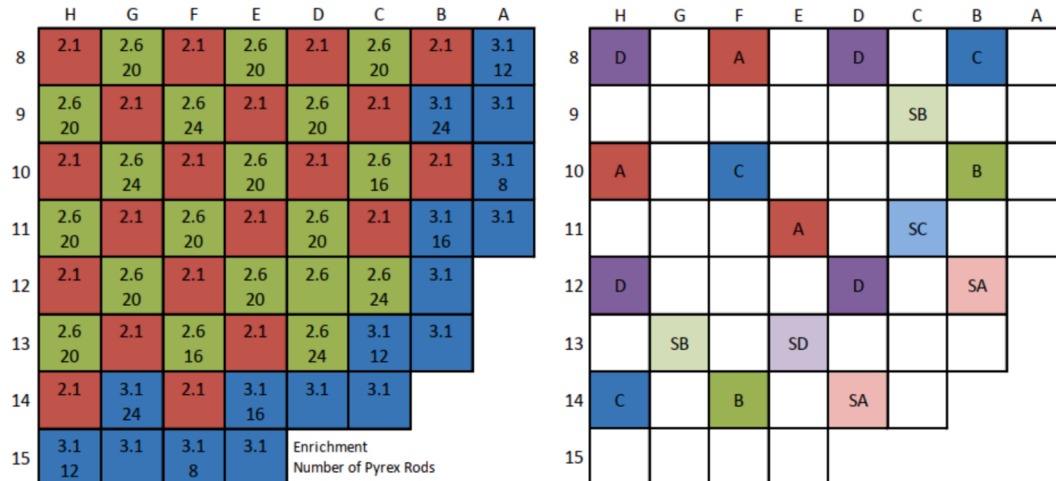


Fig. 7. Radial layout of VERA#7 benchmark.

Table 4
Operation conditions of VERA#7 benchmark.

Parameter	Value
Fuel enrichments	2.1 % 2.619 % (with 0/16/20 Pyrex burnable absorbers) 3.1 % (with 0/8/12/24 Pyrex burnable absorbers)
Reactor power	3411 MW
Inlet coolant temperature	565 K
Reactor pressure	15.5133 MPa
Inlet coolant mass flow	16590.936 kg/s
Bank D Position	215

For the large-scale whole-core VERA#7 problem, the direct Picard iteration without acceleration or a relaxation factor of 0.3 reached the maximum number of transport fission source iterations and diverged.

Table 5
Iteration performance for VERA#7 benchmark.

		Number of global iterations	Number of transport fission source iterations	Calculation time/s
Original		Diverge	Diverge	—
Relaxation	factor = 0.3	Diverge	Diverge	—
	0.5	45	171	514.184
	0.7	14	59	176.876
Nu-AA	$m_k = 1$	8	38	116.186
	2	9	41	126.989
	4	8	37	114.414
	6	8	37	111.869
	8	8	37	111.358
Nu(CV)-AA	$m_k = 1$	9	44	134.388
	2	11	50	149.383
	4	9	44	136.362
	6	9	44	133.193
	8	9	44	134.497
TH-AA	$m_k = 1$	30	80	275.411
	2	25	74	238.462
	4	23	68	224.076

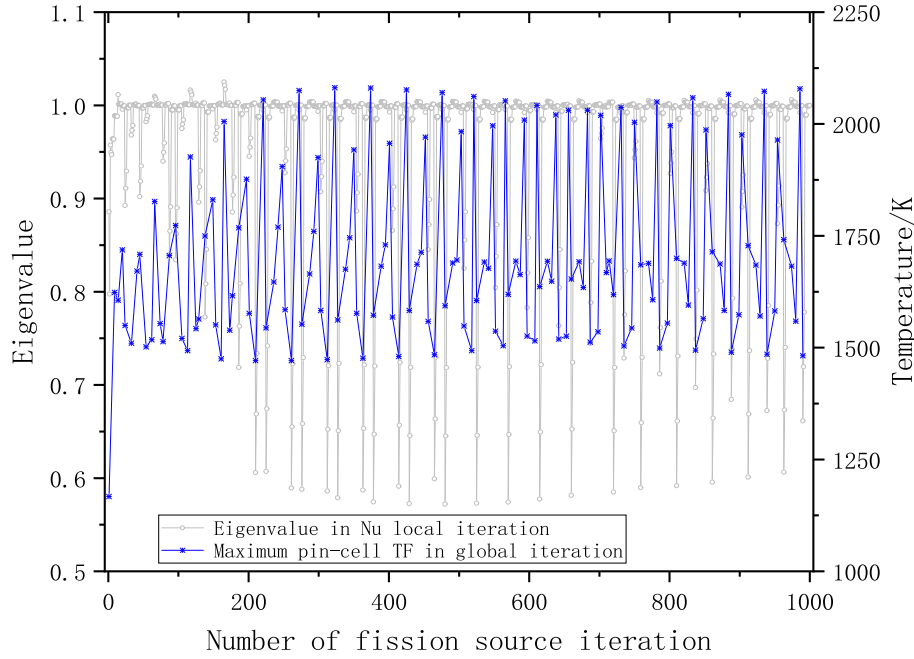


Fig. 8. k_{eff} and maximum pin-cell fuel temperature under relaxation factor = 0.3 for VERA#7 benchmark.

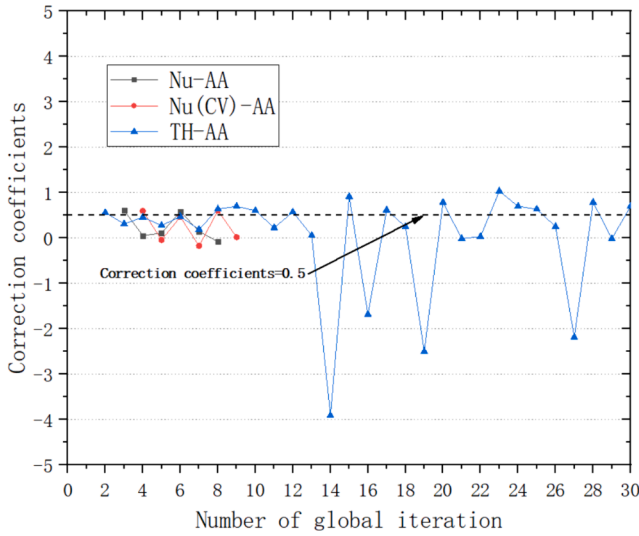


Fig. 9. Correction coefficients during the iterations for various AA schemes with $m_k = 1$ for VERA#7 benchmark.

The iteration process with a relaxation factor of 0.3 is shown in Fig. 8. In addition to the eigenvalue, which is adjusted to 1.0 between different global iterations, the maximum fuel temperature varying with global iterations is also depicted to characterize the calculation process. It can be observed that there are significant numerical oscillations during the iteration process, indicating no possibility of convergence, and increasing the number of fission source iterations will not improve the convergence situation.

In comparison, a relaxation factor of both 0.5 and 0.7 achieves computational convergence, but their convergence speed is inferior to the Nu-AA and the Nu(CV)-AA. For both of the Nu-AA and the Nu(CV)-AA, when the acceleration depth extends to 4, the total number of global iterations remains constant, converging to 8 and 9, respectively. The approximation of the direct employment of the single neutron-transport solution vector ($\mathbf{u}_{\text{tran}}^{k+1}$), $k = 0, 1, \dots$ for the Nu-AA in this transport/TH/criticality/micro-depletion coupling scenario is acceptable, which

Table 6

Iteration performance of Nu-AA-1 with different initial boron concentration for VERA#7 benchmark.

Initial boron concentration/ppm	Critical boron concentration/ppm	Number of global iterations	Number of transport fission source iterations	Calculation time/s
400	850.36	10	50	142.409
900	850.39	8	47	134.586
1285	850.42	8	38	116.186
1500	850.41	8	38	115.351

Table 7

Comparison of VERA#7 benchmark.

	CBC /ppm	Bias of CBC /ppm	RMS of assembly-power distribution bias	Max. assembly-power distribution bias
MPACT/CTF (VERA)	854	—	—	—
RMC/CTF	848	−6	1.30 %	2.40 %
MCS/TH1D	857	3	0.79 %	−1.69 %
NECP-Bamboo2.0	850	−4	0.37 %	0.69 %

can achieve the same level of the acceleration effect compared to Nu (CV)-AA with less memory required. Compared to the traditional calculation with a relaxation factor of 0.5, the total number of global iterations for Nu-AA can be reduced by 80 %. The total calculation time with Nu-AA-4 for the entire core (193 assemblies) is about 111 s (approximately 2.5 CPU-hours) as shown in Table 5, which could reduce 78 % the computation time of the calculation with a relaxation factor of 0.5. The numerical results also indicate that using the TH-only solution vector could achieve convergence but is less efficient than the AA strategy based on the neutron-transport solution vector, aligning with the theoretical expectations in section 3.3.2.

The correction coefficient values during the iterations for the various AA schemes with $m_k = 1$ for the VERA#7 benchmark are also illustrated in Fig. 9. Firstly, the fluctuation of the correction coefficient values for

1.1176	1.0326	1.1159	1.0590	1.1574	1.0547	1.0453	0.7550
1.1135	1.0315	1.1115	1.0573	1.1530	1.0555	1.0467	0.7597
-0.37%	-0.10%	-0.39%	-0.16%	-0.38%	0.08%	0.13%	0.62%
1.0326	1.1090	0.9861	1.1489	1.0826	1.1547	1.0121	0.8547
1.0315	1.1047	0.9858	1.1434	1.0813	1.1523	1.0170	0.8580
-0.10%	-0.39%	-0.03%	-0.48%	-0.12%	-0.21%	0.48%	0.38%
1.1159	0.9861	1.1327	1.0777	1.1856	1.1225	1.0517	0.7632
1.1115	0.9858	1.1277	1.0756	1.1805	1.1229	1.0539	0.7685
-0.39%	-0.03%	-0.44%	-0.19%	-0.43%	0.04%	0.21%	0.69%
1.0590	1.1489	1.0777	1.1810	1.0799	1.1165	0.9854	0.6322
1.0573	1.1434	1.0756	1.1752	1.0792	1.1156	0.9894	0.6353
-0.16%	-0.48%	-0.19%	-0.49%	-0.07%	-0.08%	0.41%	0.48%
1.1574	1.0826	1.1856	1.0799	1.2379	0.8640	0.8896	
1.1530	1.0813	1.1805	1.0792	1.2336	0.8691	0.8905	
-0.38%	-0.12%	-0.43%	-0.07%	-0.34%	0.59%	0.10%	
1.0547	1.1547	1.1225	1.1165	0.8640	0.8631	0.6060	
1.0555	1.1523	1.1229	1.1156	0.8691	0.8679	0.6091	
0.08%	-0.21%	0.04%	-0.08%	0.59%	0.56%	0.52%	
1.0453	1.0121	1.0517	0.9854	0.8896	0.6060		
1.0467	1.0170	1.0539	0.9894	0.8905	0.6091		
0.13%	0.48%	0.21%	0.41%	0.10%	0.52%		
0.7550	0.8548	0.7632	0.6322			VERA	
0.7597	0.8580	0.7685	0.6353			Bamboo2.0/START	
0.62%	0.37%	0.69%	0.48%			Deviation	

Fig. 10. Bias of assembly-power distribution for VERA#7 benchmark.

TH-AA becomes pronounced when approaching convergence in the later stages of iteration, as the reflection of the instability of the AA system. The correction coefficient values for the Nu-AA and the Nu(CV)-AA in the whole iteration process are mostly maintained within the range of (0,0.7), also close to the range in VERA#6 calculation.

For the VERA#7 benchmark, the influence of different initial conditions is also investigated. Considering that the inlet moderator temperature is given, and the initial nuclide densities are usually set to be 0, hence the different values of initial boron concentration are used for the evaluation. As mentioned above, the initial boron concentration is 1285 ppm for the calculation in Table 5. The iteration results of Nu-AA-1 with 3 more different initial boron concentration, i.e., 400 ppm, 900 ppm, and 1500 ppm, are demonstrated in Table 6. Different initial conditions have an influence on the number of global iterations and the number of transport fission source iterations. However, for Nu-AA-1, all the calculations can converge within 10 global iterations, and the deviations of final critical boron concentration between different initial conditions are less than 1 ppm.

The CBC and the assembly-power distribution results are shown in Table 7 and Fig. 10. The differences in CBC between NECP-Bamboo2.0 and other high-fidelity codes are comparable, all less than 10 ppm. Compared to the MPACT/CTF reference result provided by the VERA benchmark, NECP-Bamboo2.0 could obtain a highly accurate assembly-power distribution, the RMS and maximum bias of assembly-power distribution are 0.37 % and 0.69 %, respectively. When compared to other publicly published results of RMC/CTF and MCS/TH1D (Ma et al., 2019; Cao et al., 2020), the accuracy of NECP-Bamboo2.0 is quite competitive.

5. Conclusions

In this paper, starting from the commonly used Picard iteration process, the optimization mechanisms and limitations of the traditional strategies are discussed, and the Anderson Acceleration is selected and integrated into the NECP-Bamboo2.0 for PWR whole-core pin-by-pin Nu-TH coupling calculations. Based on the general theory of AA, a specific solution scheme suitable for pin-by-pin core calculations is provided, along with the code development of the corresponding solver, which efficiently utilizes the computing resources within a parallel environment, achieving problem resolution with minimal communication costs.

For the NECP-Bamboo2.0 tightly coupling strategies, scenarios ranging from the simple transport/TH coupling to the complex transport/TH/criticality/micro-depletion coupling are abstracted and analyzed. It compares the applicability of AA schemes based on different solution vectors and proposes that the strategy based on neutron-transport vectors is more suitable for practical physics analysis than the classical TH solution vectors. The convergence characteristics of relaxation, Nu-AA, and TH-AA in different scenarios are theoretically predicted. The correctness of the theoretical analysis and the efficiency of the AA are verified through calculations of the VERA#6 and VERA#7 problems. For the 3D whole-core Nu-TH coupling issues as VERA#7, compared to the classical Picard iteration with a relaxation factor of 0.5, the AA based on neutron physics vectors can reduce the number of global iterations by approximately 80 %, and the same level of calculation time could also be reduced.

CRedit authorship contribution statement

Sicheng Wang: Writing – original draft, Software, Methodology, Investigation. **Yunzhao Li:** Funding acquisition, Formal analysis. **Liangzhi Cao:** Writing – review & editing, Conceptualization. **Lei Chen:** Data curation, Formal analysis. **Hongchun Wu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yunzhao Li reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work is financially supported by the National Science Foundation of China (No. 11975181).

References

- Anderson, D.G.M., 2019. Comments on “Anderson acceleration, mixing and extrapolation”. *Numer. Algorithms* 80, 135–234.
- Aviles, N.A., Kelly, D.A., Aumiller, D.L., Gill, D., Siebert, B., Godfrey, A.T., Collins, B.S., Salko, R.K. 2016. Coupled MC21 and COBRA-IE solution to VERA core physics benchmark problem#6, In: PHYSOR2016, Sun Valley, ID, May 1-5.
- Basualdo, J.R., Sanchez-Espinoza, V., Stieglitz, R., Macian-Juan, R., 2020. Integration of the subchannel thermal-hydraulic code SubChanFlow into the reactor dynamics code PARCS: Development and testing based on a computational benchmark. *Prog. Nucl. Energy* 119, 103138.
- Cao, N.T.D., Hyunsuk, L., Sooyoung, C., Lee, D., 2020. MCS/TH1D analysis of VERA whole-core multi-cycle depletion problems. *Ann. Nucl. Energy* 139, 107271.
- Chaudri, K.S., Kim, J., Kim, Y., 2019. Development and validation of a fast sub-channel code for LWR multi-physics analyses. *Nucl. Eng. Technol.* 51, 1218–1230.
- Daeubler, M., Trost, N., Jimenez, J., Sanchez, V., Stieglitz, R., Macian-Juan, R., 2015. Static and transient pin-by-pin simulations of a full PWR core with the extended coupled code system DYN SUB. *Ann. Nucl. Energy* 84, 31–44.
- Daniel, J., Kelly, I.L.I., Kelly, A.E., Aviles, B.N., Godfrey, A.T., Salko, R.K., Collins, B.S., 2017. MC21/CTF and VERA multiphysics solutions to VERA core physics benchmark progression problems 6 and 7. *Nucl. Eng. Technol.* 49, 1326–1338.
- Facchini, A., Lee, J., Joo, H.G., 2021. Investigation of Anderson acceleration in neutronics-thermal hydraulics coupled direct whole core calculation. *Ann. Nucl. Energy* 153, 108042.
- Ferraro, D., García, M., Valtavirta, V., Imke, U., Tuominen, R., Leppänen, J., Sanchez-Espinoza, V. 2020. Serpent/SUBCHANFLOW pin-by-pin coupled transient calculations for a PWR minicore, *Annals of Nuclear Energy*, 137: 107090.

- Ferraro, D., Garcia, M., Imke, U., Valtavirta, V., Leppänen, J., Sanchez-Espinoza, V., 2019. Serpent/SCF pin-level multiphysics solutions for the VERA fuel assembly benchmark. *Ann. Nucl. Energy* 128, 102–114.
- García-Herranza, N., Cuervo, D., Sabatera, A., Rucabado, G., Sánchez-Cervera, S., Castro, E., 2017. Multiscale neutronics/thermal-hydraulics coupling with COBAYA4 code for pin-by-pin PWR transient analysis. *Nucl. Eng. Des.* 321, 38–47.
- Gómez-Torres, A.M., Sanchez-Espinoza, V.H., Ivanov, K., Macian-Juan, R., 2012. DYN SUB: a high fidelity coupled code system for the evaluation of local safety parameters – part I: development, implementation, and verification. *Ann. Nucl. Energy* 48, 108–122.
- Hamilton, S., Berrill, M., Clarno, K., Pawlowski, R., Alex Toth, C.T., Kelley, T.E., Philip, B., 2016. An assessment of coupling algorithms for nuclear reactor core physics simulations. *J. Comput. Phys.* 311, 241–257.
- Jbilou, K., Sadok, H., 2000. Vector extrapolation methods. applications and numerical comparison. *J. Comput. Appl. Math.* 122 (1–2), 149–165.
- Kochunas, B., Collins, B., Stimpson, S., Salko, R., Jabaay, D., Graham, A., Liu, Y., Kim, K. S., Wieselquist, W., Godfrey, A., Clarno, K., Palmtag, S., Downar, T., Gehin, J., 2017. VERA core simulator methodology for PWR cycle depletion. *Nucl. Sci. Eng.* 185, 217–231.
- Lee, J., Joo, H.G., 2022. Convergence analysis of fixed-point iteration with Anderson Acceleration on a simplified neutronics/thermal-hydraulics system. *Nucl. Eng. Technol.* 54, 532–545.
- Li, Y., Yang, W., Wang, S., Hongchun, Wu., Cao, L., 2020. A three-dimensional PWR-core pin-by-pin analysis code NECP-Bamboo2.0. *Ann. Nucl. Energy* 144, 107507.
- Ma, Y., Liu, S., Luo, Z., Huang, S., Li, K., Wang, K., Ganglin, Yu., Hongxing, Yu., 2019. RMC/CTF multiphysics solutions to VERA core physics benchmark problem 9. *Ann. Nucl. Energy* 133, 837–852.
- Müller, E., 2022. Mitigating the numerical xenon instability in deterministic reactor burnup calculations. *Ann. Nucl. Energy* 174, 109141.
- Park, H.J., Kim, S.J., Kwon, H., Cho, J.Y., 2020. BEAVRS benchmark analyses by DeCART stand-alone calculations and comparison with DeCART/MATRA multiphysics coupling calculations. *Nucl. Eng. Technol.* 52, 1896–1906.
- Ragusa, J.C., Mahadevan, V.S., 2009. Consistent and accurate schemes for coupled neutronics thermal-hydraulics reactor analysis. *Nucl. Eng. Des.* 239, 566–579.
- Ramos, E., Roman, J.E., Abarca, A., Miró, R., Bermejo, J.A., 2016. Control rod drop transient analysis with the coupled parallel code pCTF-PARCSv2.7. *Ann. Nucl. Energy* 87, 308–317.
- Rempe, K.R., Smith, K.S., Henry, A.F., 1989. SIMULATE-3 pin power reconstruction: methodology and benchmarking. *Nucl. Sci. Eng.* 103, 334–342.
- Saad, Y., Schultz, M.H., 1986. GMRES: A generalized minimal residual algorithm for solving nonsymmetric linear systems. *SIAM J. Sci. Stat. Comput.* 7 (3), 856–869.
- Senecal, J.P., Ji, W., 2017. Approaches for mitigating over-solving in multiphysics simulations. *Int. J. Numer. Meth. Eng.* 112, 503–528.
- Senecal, J.P., Ji, W., 2018. Development of an efficient tightly coupled method for multiphysics reactor transient analysis. *Prog. Nucl. Energy* 103, 33–44.
- Toth, A., Kelley, C.T., 2015. Convergence analysis for Anderson acceleration. *SIAM J. Numer. Anal.* 53 (2), 805–819.
- Walker, H.F., Ni, P., 2011. Anderson acceleration for fixed-point iterations. *SIAM J. Numer. Anal.* 49 (4), 1715–1735.
- Wang, S., Li, Y., Cao, L., Hongchun, Wu., 2023. A cross-section correction model for history effect treatment in pin-by-pin code NECP-Bamboo2.0. *Ann. Nucl. Energy* 194, 110088.
- Wang, S., Kim, Y., Li, Y., Cao, L., 2024. An embedded coupling design and development of NECP-Bamboo2.0 and START for PWR whole-core pin-by-pin analysis. *Ann. Nucl. Energy* 199, 110353.
- Wang, J., Wang, Q., Ding, M., 2020. Review on neutronic/thermal-hydraulic coupling simulation methods for nuclear reactor analysis. *Ann. Nucl. Energy* 137, 107165.
- Yamamoto, A., Ikeno, T., 2005. Impact of pin-by-pin thermal-hydraulic feedback modeling on steady-state core characteristics. *Nucl. Technol.* 149 (2), 175–188.
- Yamamoto, A., Kitamura, Y., Yamane, Y., 2003. Cell homogenization methods for pin-by-pin core calculations tested in slab geometry. *Ann. Nucl. Energy* 31, 825–847.
- Yang, W., Hongchun, Wu., Li, Y., Cao, L., Wang, S., 2018a. Acceleration of the exponential function expansion nodal SP3 method by multi-group GMRES algorithm for PWR pin-by-pin calculation. *Ann. Nucl. Energy* 120, 869–879.
- Yang, W., Hongchun, Wu., Li, Y., Yang, J., Cao, L., 2018b. Development and verification of PWR-core fuel management calculation code system NECP-Bamboo: Part II Bamboo-Core. *Nucl. Eng. Des.* 337, 279–290.
- Yang, D., Liu, X., Tengfei Zhang, Xu., Cheng., 2020. A comparison of three algorithms applied in thermal-hydraulics and neutronics codes coupling for lbe-cooled fast reactor. *Ann. Nucl. Energy* 149, 107789.
- Zerkak, O., Kozłowski, T., Gajev, I., 2015. Review of multi-physics temporal coupling methods for analysis of nuclear reactors. *Ann. Nucl. Energy* 84, 225–233.
- Zhang, B., Hongchun, Wu., Li, Y., Cao, L., Shen, W., 2016. Evaluation of Pin-cell homogenization techniques for PWR Pin-by-pin calculation. *Nucl. Sci. Eng.* 186 (2), 134–146.
- R. Montgomery, 2011. TVA reported core performance information for Watts Bar Unit 1 Cycle 1, CASL, CASL-I-2012-0101-000-b.